

Implications of RNA virus persistence for post-acute sequelae and chronic inflammatory syndromes

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Viral persistence refers to the ability of a virus to remain in its host for an extended period. Although most non-retroviral RNA viruses are traditionally known for causing acute, self-limiting infections, accumulating evidence of the detection of viral products long after the infectious virus is cleared suggests that RNA viruses can establish persistent infections. Persistent viral products include replication-competent viral genomes, viral proteins, mutated viruses or non-standard viral genomes. The persistence of viral products entails a continuous interaction with the host and is often associated with prolonged tissue inflammation and chronic disease. Here, we discuss emerging evidence of the persistence of viral products from viruses commonly considered to cause acute infections. We explore the known requirements for virus persistence and the implications for the development of post-acute sequelae and predisposition to chronic inflammatory syndromes.

RNA viruses include some of the most common human viruses, such as seasonal respiratory syncytial virus (RSV), coronaviruses (CoV), human influenza viruses and human parainfluenza viruses (HPIVs); some of the most infectious viruses, such as measles virus (MeV); some of the most debilitating viruses, such as chikungunya (CHIKV); and some of the most lethal viruses, such as Ebola virus (EBOV) and the henipaviruses. Except for retroviruses, hepatitis C virus (HCV) and arenaviruses, RNA viruses are generally thought to cause acute infections that resolve quickly, leading to the development of protective immunity. However, recent technological advances reveal that several RNA viruses or their viral products can persist long term in the host, impacting health for months or years after initial infection^{1–9}. Although evidence of transmission of persistent non-retroviral RNA viruses in humans remains limited, compelling examples do exist. For example, several reports have shown that EBOV can be transmitted from the male reproductive tract to sexual partners months after infection^{10–13}. Evidence for the persistence of HPIVs in humans dates back to the early 1980s, when HPIVs were detected in infection outbreaks among individuals isolated for several months in Antarctica¹⁴. More recently, persistent HPIV shedding has been documented in individuals who are immunosuppressed, even when asymptomatic¹⁵.

Furthermore, increasing evidence indicates that persistent viral products can lead to sustained tissue inflammation and chronic diseases. This is likely due to the presence of pathogen-associated molecular patterns that trigger the host's innate immune response^{2,16–18}. Examples include MeV infections and the onset of subacute sclerosing panencephalitis (SSPE)^{18–22}, SARS-CoV-2 infection and the development of long COVID^{5,6,17,23}, CHIKV infection and the onset of persistent myalgia^{7,24–26} and EBOV infection and post-EBOV disease^{8,27–29}. Demonstrating a causal relationship between the persistence of viral products and the development of chronic disease in humans is challenging owing to the need for isolated populations, the inaccessibility of persistently infected cells in live individuals and ethical issues with human experimentation. However, a more conclusive causal relationship was recently demonstrated for parainfluenza viruses (PIVs) and CHIKV. Using the murine PIV Sendai virus (SeV), our group identified lung macrophages, type 2 innate lymphoid cells (ILC2s) and dendritic cells (DCs) containing viral RNA and protein long after the acute infection actively produced proinflammatory molecules. Importantly, eliminating these cells drastically reduced chronic lung inflammation². For CHIKV, recent evidence in mice showed that inflammatory macrophages accumulate in joint-associated tissues and harbor viral

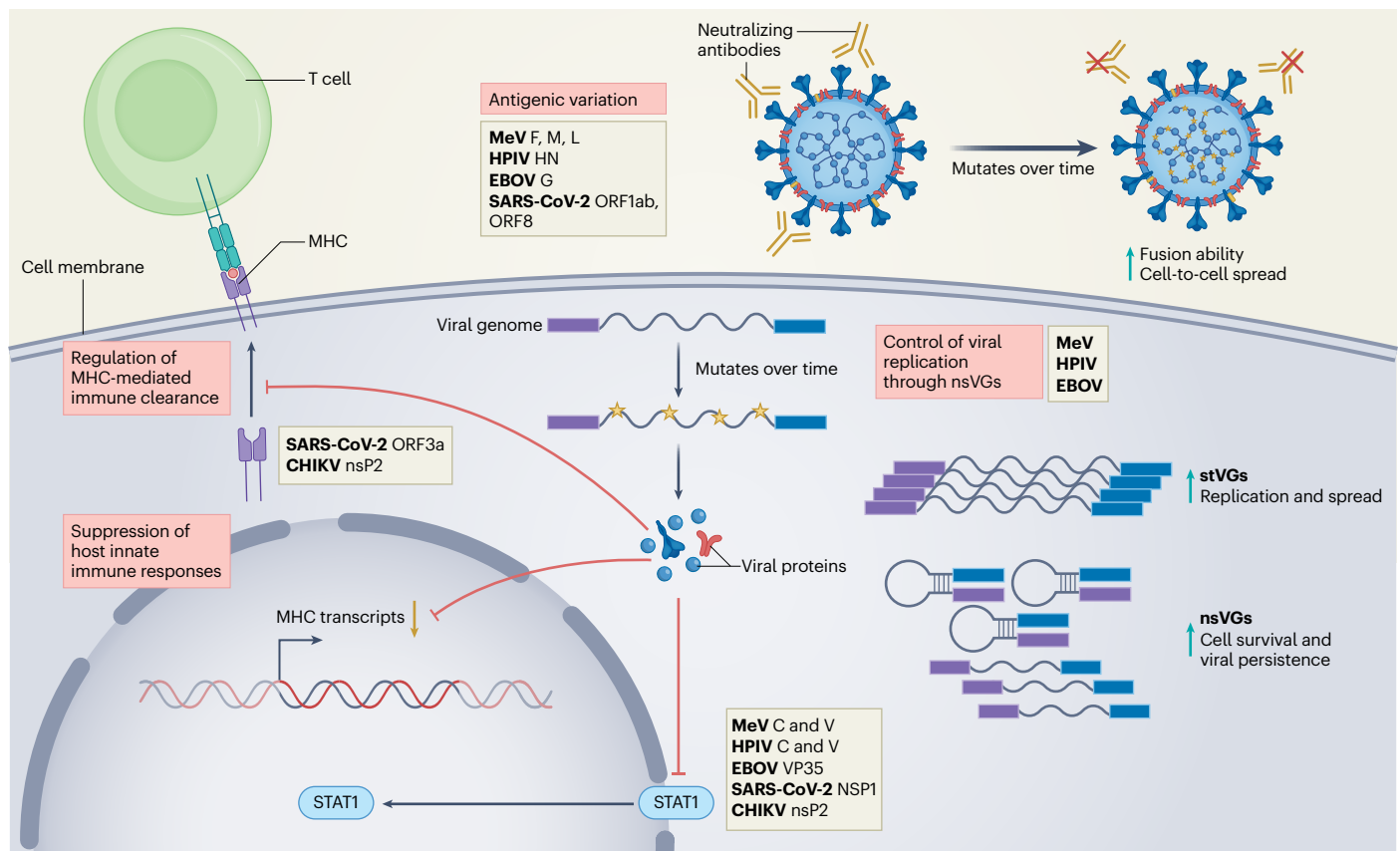


Fig. 1 | Mechanisms of RNA virus persistence: immune evasion strategies.

RNA viruses use diverse strategies to evade immune detection and establish persistence. These include (1) antigenic variation (acquisition of advantageous mutations that enable the virus to escape immune recognition), (2) production of nsVGs (generation of truncated viral genomes that reduce overall viral replication), (3) suppression of IFN production or signaling (disruption of

IFN signaling pathways, thereby inhibiting the activation of downstream effectors essential for the antiviral response), and (4) inhibition of antigen presentation (interference with the expression of MHC to prevent effective antigen presentation to immune cells); F, fusion; G, glycoprotein; HN, hemagglutinin-neuraminidase; L, large protein (polymerase); M, matrix; NSP1, non-structural protein 1; stVGs, standard viral genomes.

RNA during chronic disease, and suggests that CD4⁺ T cells infiltrating these tissues have a role in inflammation⁹.

Persistent infections require immune evasion or subversion, allowing continuous virus replication and/or long-term retention of viral material. Because all RNA viruses face robust immune surveillance, their persistence in the host may result from evolved strategies to evade or suppress the host's innate immune defenses or disrupt antigen presentation, which is essential for the development of cytotoxic effector cells that clear the infection. Infection of cellular niches that offer favorable conditions, such as longevity, location or impaired antiviral signaling, can also contribute to virus persistence. Here, we review advances in our understanding of the mechanisms driving the persistence of viral products and viruses traditionally considered to cause acute infections, as well as emerging evidence linking persistent viral products to post-acute sequelae and predisposition to chronic inflammatory syndromes. The mechanisms driving persistence of the most classical and well-studied RNA viruses that cause chronic infections, such as HCV and arenaviruses, have been recently reviewed elsewhere^{30,31}. Instead, this review focuses on emerging evidence of the persistence of viruses and viral products traditionally considered to cause acute infections.

Immune mechanisms of RNA virus persistence

DNA viruses use genetic strategies to persist, including maintaining their genomes as episomes in the nucleus of dividing cells (for example, herpesviruses and papillomaviruses) or integrating their genome into the host genome. Retroviruses, such as HIV, integrate a cDNA version of

their genome into the host genome and have a high viral mutation rate that facilitates evasion of neutralizing antibodies. Persistence of other RNA viruses relies mainly on evading or modulating the host immune response, and, in some cases (for example, mumps virus and PIVs), persistence has been associated with the formation of viral condensates that presumably protect the virus from immune recognition^{32,33}. The critical role of the immune response in preventing the persistence of acute RNA viruses is evident in individuals who are immunocompromised in response to specific treatments, in individuals with congenital immunodeficiency and in individuals experiencing age-related immune decline, conditions where viral replication is prolonged with an increased risk of developing persistent infections^{34–38}. Adaptive immune mechanisms that support the maintenance of persistent infections, such as T cell exhaustion and suppression of T cell responses, have been thoroughly covered in recent reviews^{31,39}. Here, we discuss virus-driven mechanisms that restrict early immune responses, enabling high replication of acute RNA viruses and contributing to virus persistence (Fig. 1).

Suppression of early host antiviral responses

The host immune response to RNA viruses is initiated when viral pathogen-associated molecular patterns are recognized by virus sensor molecules that drive the expression of type I and III interferons (IFNs), as well as cytokines and chemokines. Type I and III IFNs signal for the expression of hundreds of genes that block virus replication. In addition, type I IFNs amplify and activate cellular responses that contribute to controlling virus spread. For example, type I IFNs are critical

for the effective function of natural killer (NK) cells that participate in early infection control⁴⁰ and for the maturation of DCs, a process that is essential to generate type I adaptive responses that clear the virus and protect from reinfection via the action of cytotoxic T cells⁴¹. RNA viruses that are adapted to their host encode proteins that directly block early steps of the antiviral response, allowing the virus to replicate and spread while limiting immune intervention. Therefore, interfering with type I and III IFN production leads to higher rates of virus replication and spread, which can facilitate the establishment of a chronic infection.

The list of viral proteins from acute RNA viruses that interfere with the type I/III IFN pathway is extensive. Some examples are the V and C proteins of MeV and HPIVs^{42–44}, non-structural protein 2 (nsP2) of CHIKV^{45,46} and viral protein 35 (VP35) of EBOV^{47–49}. SARS-CoV-2 allocates over one-third of its genome to encode proteins that antagonize almost every level of the IFN response⁵⁰. Viral-encoded antagonists can interfere at multiple levels within the IFN signaling cascade by inhibiting sensing by RIG-I-like receptors or other specialized cellular proteins, blocking signaling molecules of the IFN induction pathway or blocking type I/III IFN signaling itself^{42–44,47–49,51}. RNA viruses can also directly or indirectly interfere with DC and NK cell function. Examples include the MeV hemagglutinin protein and EBOV VP35 that inhibit maturation and activity of DCs^{52–54}.

Evidence suggests a causal link between early immune response inhibition by viruses, high viral load during the acute infection phase and the establishment of persistent infections. For example, CHIKV persistence in the spleen of aged rhesus monkeys is linked to reduced innate and adaptive immune responses⁴⁵. MeV creates an immunosuppressed environment by decreasing the numbers of T cells and B cells in circulation, thereby facilitating the establishment of persistence in lymphoid tissues and immune-privileged sites like the brain^{55,56}. SARS-CoV-2 inhibition of the type I IFN response and associated DC dysfunction observed in severe acute cases is linked to systemic viral spread. This dysfunction persists for several months, likely contributing to the long-term presence of SARS-CoV-2 antigens in various tissues, including the intestines and brain^{57–61}. In addition, immune-escape mutations in the SARS-CoV-2 spike protein and deletions in *ORF1ab* and *ORF8*, accessory genes with immune-modulatory functions, are frequently detected in hosts with persistent infection³. Last, HPIV shedding is often observed in individuals who are immunosuppressed, particularly in recipients of hematopoietic cell transplants^{35,62}. These examples demonstrate that failure to mount an effective antiviral response early in infection results in increased virus replication that directly or indirectly creates permissive conditions for the establishment of viral persistence.

Regulation of MHC expression and function

Another strategy for RNA viruses to suppress the establishment of an effective antiviral immune response is to disrupt the expression of major histocompatibility complex (MHC) molecules and/or the presentation of viral antigens on the surface of infected cells. These processes are essential for the killing of infected cells by cytotoxic T cells. Viruses can downregulate the expression of MHC molecules both at the transcriptional and post-translational levels. One example is during HCV infection, where the virus core protein induces the degradation of MHC class I (MHCI), impairing the presentation of viral antigens to cytotoxic T lymphocytes and allowing the virus to evade clearance by the immune system⁶³. SARS-CoV-2 also disrupts MHCI expression in humans, cultured cells and mice by targeting IFN signaling^{64,65}. In addition, the SARS-CoV-2 ORF3a accessory protein interferes with MHCI intracellular trafficking, and the ORF7a protein prevents proper MHCI assembly⁶⁶. Similarly, CHIKV nsP2 disrupts MHCI antigen presentation, allowing the virus to escape immune recognition by T cells⁶⁷. Although not all RNA viruses that exploit this pathway are proven to persist, hijacking MHC antigen presentation is a strategy that impairs downstream activation of effector immune

cells. This facilitates the survival of infected cells that serve as reservoirs for virus persistence.

Viral mechanisms of RNA virus persistence

After establishing infection in potential viral reservoirs, many RNA viruses acquire adaptive mutations that reduce their replication rate, allowing the infected cell to evade immune detection and facilitating virus persistence. In some persistent infections, mutations that favor cell-to-cell spread over virus particle production, together with antigenic variation, allow the virus to escape antibody detection²². In addition, most, if not all, RNA viruses generate defective or truncated versions of the viral genome known as non-standard viral genomes (nsVGs) or defective viral genomes that interfere with virus replication and can also minimize the exposure of viral antigens to the immune system. nsVGs arise during viral replication and interfere with replication of the standard virus by several mechanisms, including competing for viral and host resources, inducing potent IFN production early after infection and regulating transcription of viral genes^{40,68–71}. Evidence since the early days of virology has associated the production of nsVGs with the establishment of persistent RNA virus infections in vitro^{72–75}, and emerging evidence supports their role in vivo. Here, we provide specific examples of these viral-driven strategies as contributors to the establishment of persistence by acute RNA viruses (Fig. 1).

Antigenic variation

Antigenic variation is a common strategy to escape antibody recognition by RNA viruses. Their high mutation rate allows them to change their external proteins enough to be undetectable by antibodies^{76,77}. Antigenic variation can also change the way the virus spreads. For example, MeV brain tropism is associated with hyperfusogenic mutations in the viral fusion (F) protein that allow for increased virus propagation via cell-to-cell fusion while reducing viral particle production^{19,78–81}. In addition, mutations in the viral matrix (M) protein typically detected in postmortem SSPE tissue are thought to impair the formation of viral particles^{18,82,83}. Mutations in the RNA-dependent RNA polymerase (L protein) that result in reduced viral replication were described in an SSPE-causing virus^{22,84}. By reducing viral protein expression and infectious particle release, MeV persists by evading immune detection and minimizing neuronal death. A similar strategy is observed in EBOV infection, as mutations in the virus glycoprotein alter viral entry dynamics and promote non-lytic infection^{85–88}. Similarly, analysis of HPIV3 persisting in individuals who are immunocompromised showed accumulation of mutations in the virus hemagglutinin-neuraminidase attachment protein that led to altered particle fusion dynamics, increased avidity to cellular receptors and preference for cell-to-cell transmission of the virus³⁵.

Generation of nsVGs

In the early 1970s, RNA virus persistence was described as the onset of an ‘asynchronous cycle’ that fluctuates between standard virus replication and the production of defective interfering particles in the infected cell⁸⁹. In this model, defective interfering particles are produced gradually during virus replication until they predominate over the standard virus. The high concentration of defective interfering particles, in turn, interferes with standard virus particle production and reduces the amount of standard virus. Once the standard virus levels are low, defective interfering particles are produced at a lower rate, allowing the standard virus to resurface and initiate the cycle again⁸⁹. A newer model for nsVG-driven persistence was generated based on the observation that during infection with SeV or RSV, a subset of cells is dominated by standard virus genomes, whereas others are dominated by nsVGs^{90–92}. The heterogeneous composition of viral genomes in these cells was associated with distinct host responses to the virus, including the switch of the TNF receptor from proapoptotic TNFR1 to prosurvival TNFR2 in nsVG-enriched cells, allowing their survival

over other infected cells and facilitating the long-term persistence of the virus *in vitro*⁹⁰.

The demonstration that nsVGs are present in human samples infected with MeV, RSV and EBOV, together with their detection in persistent infections, support the hypothesis that nsVGs facilitate the establishment of RNA virus persistence during natural infections^{40,93–95}. Analysis of brain tissue from an individual who succumbed to SSPE two decades after MeV infection identified nsVGs across multiple brain regions⁹³. In studies of EBOV infection in rhesus macaques, nsVGs were detected in the serum and testes of infected animals at 5, 7 and 9 days after infection, and nsVGs have also been detected in semen from men with persistent EBOV infection^{4,95}. Although direct evidence of an *in vivo* role for nsVGs in prolonging the duration of RNA virus infections is lacking, their ubiquitous presence across RNA viruses suggests that nsVG generation is a conserved proviral strategy. We propose that by stimulating the host IFN response, promoting cell survival mechanisms and reducing viral RNA and protein, nsVGs act as a switch between high and low viral replication, enabling the virus to evade immunity and extend its presence in the host⁹⁶.

Establishment of cellular reservoirs of viral products

To persist in immunocompetent hosts, RNA viruses rely on finding a niche that is or can become resistant to immune clearance. Although cellular reservoirs have been well studied for retroviruses such as HIV-1, our understanding of such reservoirs for other RNA viruses remains limited⁹⁷. Immune-privileged organs, such as the brain and testes, long-lived cells and cells with low expression of sensor molecules that trigger antiviral immunity in response to virus recognition (for example, RIG-I-like receptors) are good candidate host reservoirs for RNA viruses^{17,98,99}. However, recent studies suggest that RNA viruses or their viral products can also persist in other cell types, including mucosal structural cells^{100,101}, sensory cells^{102,103}, muscle cells^{24,25}, synovial joints¹⁰⁴, macrophages^{2,9,105,106}, DCs and ILC2s², significantly expanding the cell types that could be long-term viral reservoirs.

RNA virus persistence in immune-privileged sites

The central nervous system is considered an immune-privileged system as it mounts limited immune responses to threats. Limited immune responses prevent swelling and damage of the tissues^{98,107}. In some cases, neurotrophic viruses, such as MeV, harness such an immunosuppressed environment to establish persistence within neurons. Autopsy studies of individuals who succumbed to SSPE confirmed the presence of MeV RNA and protein in the brain^{22,93,108,109}. Interestingly, MeV can infect and persist in neurons and astrocytes that lack the MeV receptors CD150 and nectin-4, indicating the use of alternative entry mechanisms or cell-to-cell transmission for the virus to establish itself in the brain^{110,111}. Similarly, EBOV has been found in the brain ventricular system of non-human primates, even after standard antiviral treatment with monoclonal antibodies^{112,113}. Moreover, onset of meningoencephalitis associated with viral persistence was reported in two survivors of EBOV infection, where the virus was detected in the blood and cerebrospinal fluid 342 days after initial infection for one individual and 137 days after initial infection for the other²⁸. SARS-CoV-2 RNA, as well as spike protein and nucleoprotein, have been reported in the skull, meninges and brain tissue of individuals who succumbed to COVID-19 and in those that died from non-COVID-19-related causes, suggesting long-term persistence of SARS-CoV-2 in the brain^{23,100}. Experiments in mice have also shown persistence of RNA viruses in the brain, including enteroviruses^{114,115} and Zika virus^{98,116}, suggesting that long-term neurological sequelae of viral infections may be in part due to the persistence of these viruses in the brain.

In the testes, physical barriers together with an anti-inflammatory microenvironment that protects sperm from autoimmunity create an immune-privileged organ where acute RNA viruses can persist.

EBOV RNA and infectious virus have been detected in the semen of male survivors up to 200 days after initial infection. Additionally, persistent virus was shown to be transmitted to female partners, leading to their subsequent infection^{10,11,29}. Similarly, Lassa virus RNA and infectious virus were recovered from the seminal fluid of survivors up to 12 months into the study¹¹⁷, and Zika virus has been detected in seminal fluid for up to 83 days after acute disease and was shown to be transmitted to female partners^{118,119}. The eye is another immune-privileged site where local immune responses are more suppressed to limit inflammation-induced impairment of vision. The eye has been identified as a site for persistent EBOV¹²⁰. These examples highlight the role of immune-privileged organs as niches for acute RNA virus persistence and emphasize the importance of studying and identifying persistent viral reservoirs to reduce the transmission of pathogenic viruses and alleviate the public health burden of post-viral chronic conditions.

RNA virus persistence in immune cells

Infection of mobile lymphocytes and myeloid cells offers an opportunity for viruses to circulate throughout an organism and reach immune-privileged sites. In addition, non-circulating immune cells such as macrophages, which can take up infected cells and are relatively long lived, can serve as reservoirs for RNA viruses (Fig. 2). Although SARS-CoV-2 infection is primarily associated with respiratory pathology, it is now recognized to infect multiple systems, including cells of the immune system^{5,17}. Studies in non-human primate models showed that replication-competent SARS-CoV-2 can persist in alveolar macrophages for over 200 days after infection¹²¹. SARS-CoV-2 proteins and replicative RNA intermediates have also been detected in circulating monocytes, B cells, T cells and pulmonary inflammatory macrophages from individuals with COVID-19 for up to 15 months after acute infection^{5,17,122}.

Recently, our group demonstrated that the mouse PIV SeV establishes long-term infection in the lung, with persistent viral products detected in alveolar macrophages, ILC2s, DCs and other immune cells². Similarly, in rhesus macaques, MeV RNA is retained in lymphoid tissues for up to 6 months, mostly in B cells^{55,56}. Moreover, clinical studies have demonstrated that children with chronic adenotonsillar disease harbor viral RNA and antigens from influenza, RSV and other viruses in tonsillar lymphocytes long after primary infection^{123–125}.

Apart from respiratory RNA viruses, arthritogenic viruses like CHIKV cause infections associated with persistent RNA in immune cells. Studies in non-human primates have identified macrophages as cellular reservoirs that contain significant amounts of CHIKV antigen and RNA for up to 3 months after infection. Inflammatory macrophages were also recently found in joint-associated tissues⁹. These studies confirm findings in humans, where synovial macrophages were found to harbor persistent CHIKV^{105,106}. Infection of motile immune cells can be a means of increased viral dissemination, commonly referred to as ‘trojan horse’-mediated dissemination. One of the most well-studied examples, Zika virus, targets human CD14⁺ monocytes in circulating blood and can be found in monocyte-derived cells in the central nervous system of human fetuses^{99,126}. Zika virus infection of monocytes is productive and alters the expression of adhesion factors that increase the likelihood of blood–brain barrier crossing by infected cells and delivery of infectious virus to the central nervous system¹²⁶. These observations highlight the complexity of virus–immune cell interactions and reinforce the need for a thorough understanding of viral persistence within immune cells.

RNA virus persistence in other cell types

Structural cells lining mucosal surfaces, such as respiratory and intestinal epithelial cells, are attractive niches for viral persistence owing to their slow turnover and accessible location. SARS-CoV-2 RNA and spike protein are detected in intestinal epithelial cells months after acute infection^{100,101}, and fecal shedding of viral RNA is detected up

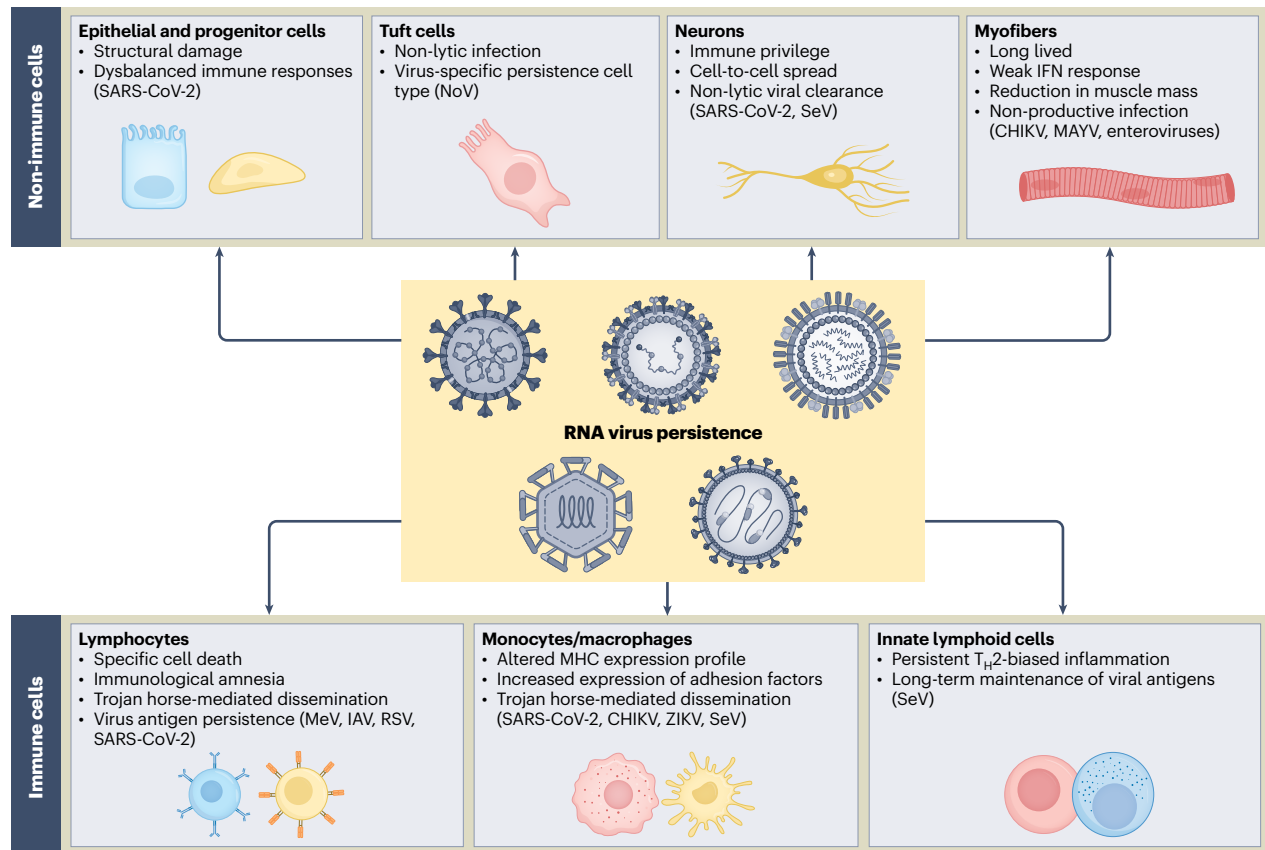


Fig. 2 | RNA virus persistence in diverse cell types. The persistence of RNA viruses, whether as fully infectious particles or viral subproducts, can significantly impact the biology of various non-immune and immune cells. This persistence can disrupt cellular functions and lead to alterations in tissue homeostasis. IAV, influenza A virus; NoV, norovirus; T_H2, type 2 helper T cell.

to 7 months after infection^{127,128}. One study reported the detection of SARS-CoV-2 spike protein-encoding RNA in colorectal tissue 676 days after initial infection¹²⁹. Owing to the potential role of persistent viral products in chronic intestinal symptoms, increasing efforts are focused on identifying the specific intestinal cell types harboring SARS-CoV-2 material with the purpose of developing targeted strategies to alleviate long COVID symptoms. Another example of an RNA virus persisting in non-immune cells in the gastrointestinal tract is norovirus, which establishes persistent infection in mouse intestinal tuft cells. These cells are chemosensory epithelial cells found in mucosal epithelia and can escape CD8⁺ T cell killing, thus they are considered an immune-privileged cell type¹⁰³.

Persistence in musculoskeletal tissue is a hallmark of arthritogenic RNA virus pathogenesis. CHIKV has long been recognized to cause infections in which viral RNA persists in musculoskeletal tissue in animal models^{24,25,130}. High levels of viral RNA are found chronically in myofibers and muscular fibroblasts of mice infected with CHIKV and Mayaro virus (MAYV), and this correlates with lingering expression of interleukin-10 and transforming growth factor- β and reduced muscle fiber mass^{25,131}. Similarly, but not restricted to arthritogenic arboviruses, enteroviruses are known to persist in the myocardium with potential links to cardiomyopathies¹³². Cardiomyopathies are frequently idiopathic, and more studies are needed to determine if persistent viral infections and/or consequent immune dysregulation are responsible for these diseases.

Sensory tissues have also been demonstrated to be niches for the persistence of RNA viruses and viral products. SARS-CoV-2 can persist in the taste papillae of humans for months after infection, specifically in basal epithelial cells and lamina propria cells. This persistence is associated with structural damage and an aberrant immune response

that contributes to long-term taste dysfunction¹⁰². Similarly, SeV products can be detected long after acute infection in long-lived olfactory neurons, providing a potential link between respiratory viral infections and post-viral anosmia^{133,134}.

Whether RNA viruses can persist in basal cells of the epithelium, which are responsible for generating most epithelial cells, is unknown. There is evidence that tissue stem cells are resistant to T cell killing, which would make them an attractive long-term home for viruses¹³⁵. Structural cells that are reservoirs for RNA viruses may experience damage or long-term changes that could impact how the tissue senses and responds to future infections. Understanding how viral persistence can alter the identity and function of non-immune cells may be key to explaining some of the lingering symptoms observed after acute viral infection and could help develop strategies to prevent or reverse chronic tissue dysfunction.

Long-term consequences of viral persistence

The prolonged detection of RNA viruses or viral products in diverse tissues suggests that many of these viruses can persist in the host for a long time after initial infection. This persistence has been associated with the development of chronic manifestations long after the resolution of the acute phase of infection. These manifestations, collectively referred to as post-acute or post-viral syndromes, challenge the long-standing understanding of acute RNA viral infections and raise questions not only about the host–viral interactions but also about the treatment and prognosis of acute infections¹³⁶.

The occurrence and severity of post-viral syndromes are diverse and may depend on the virus involved (Table 1). The emergence of long COVID, also referred to as post-acute sequelae of SARS-CoV-2 infection, suggests that viral persistence may cause a wide range of symptoms,

Table 1 | Detection of viral products in tissue reservoirs associated with post-acute viral syndromes

Post-acute viral syndrome	Location of viral persistence	Viral product and method of detection	Associated symptoms
Post-acute sequelae of SARS-CoV-2, long COVID, long-haul COVID or persistent post-COVID syndrome	Enterocytes ¹⁰¹ Taste buds ¹⁰² Skull–meninges–brain axis ²³ Non-classical monocytes ¹⁷ T cells and B cells ¹²²	SARS-CoV-2: nucleoprotein, spike and ORF1ab detected by immunostaining and/or viral RNA (RT–qPCR)	Extreme tiredness Memory problems (brain fog) Lightheadedness or dizziness Gastrointestinal problems Problems with taste or smell ^{13,5,127,128,149}
SSPE	Neurons ⁸¹ Astrocytes ²²	MeV: full genome from SSPE tissue and viral mRNA detected by Northern blotting and RNA-seq; viral N protein detected by immunohistochemistry; nsVGs detected by RNA-seq	Neurological: personality or behavior changes, irritability, dementia, lethargy, intellectual deterioration; focal or generalized seizures, with one-third of individuals developing epilepsy Musculoskeletal: decline in motor function, myoclonus (jerking, involuntary movements), dyskinesia and dystonia ¹⁵⁰
Post-EBOV disease syndrome	Brain ventricles ¹¹³ CSF ²⁸ Blood ²⁸ Aqueous humor (eye) ¹²⁰ Semen ¹¹	EBOV: full genome and viral mRNA detected by RT–qPCR and next-generation sequencing; infectious virus detected by culture of aqueous humor	Neurological: headaches, sleep disturbances, short-term memory impairment, dysesthesias, mood disorders (depression, anxiety) Respiratory: cough, shortness of breath (dyspnea) Musculoskeletal: myalgias (muscle pain), arthralgias (joint pain) Cardiovascular: cardiac manifestations (palpitations) Gastrointestinal: abdominal pain Dermatological: cutaneous rash, alopecia (hair loss) Ophthalmic/auditory: blurred vision, uveitis, hearing loss Genitourinary: erectile dysfunction ^{8,27,139}
Post-CHIKV disease or chronic fatigue syndrome/myalgic encephalomyelitis	Macrophages ¹⁰⁵ Myofibers ²⁵ Dermal and muscle fibroblasts ²⁵	CHIKV: viral RNA detected by RT–qPCR; viral protein detected by immunofluorescence	Musculoskeletal diseases: persistent arthralgia, peripheral and large synovial joint comorbidities, arthritis Other: alopecia, depression, mood swings, sleep disturbances, blurred vision, memory loss ^{7,26}
Post-HPIV asthma-like disease	ILC2s, tissue macrophages, DCs ²	Murine PIV SeV: viral RNA detected by RT–qPCR; viral protein detected by immunofluorescence	Lung disease showing alveolitis, bronchiolization of the alveolar compartment, vasculitis and bronchoalveolar lymphoid tissue formation

CSF, cerebrospinal fluid; RNA-seq, RNA sequencing; RT–qPCR, quantitative PCR with reverse transcription.

including fatigue, cognitive dysfunction, gastrointestinal symptoms, cardiovascular complications and taste and smell dysfunction that persist for weeks or months after infection^{102,127,137,138}. The detection of SARS-CoV-2 RNA and viral protein in various tissues, such as the gut, brain and lymph nodes, months after infection demonstrates incomplete clearance of viral products by the host immune system¹⁰⁰. For example, prolonged shedding of SARS-CoV-2 RNA in the stool is associated with gastrointestinal symptoms including abdominal pain, nausea and vomiting^{101,127,128}. Another example is the reported persistence of SARS-CoV-2 spike protein in basal epithelial cells of taste papillae, which is linked to long-term taste dysfunction¹⁰², while persistent RNA in the brain is associated with neurological impairment¹⁰⁰.

EBOV persistence can also drive systemic and delayed complications affecting multiple organs^{139–141}. In some cases, post-EBOV disease results from reactivation of the virus that had been hiding from the immune system for months after the primary infection. For example, EBOV was recovered from the eye of a survivor who suffered from sudden uveitis, a chronic inflammation in the eye that can lead to vision loss^{27,120}. Post-viral persistence has also been associated with fatal outcomes, such as the development of SSPE years after initial MeV infection. SSPE cases are often observed at higher rates among unvaccinated children, and individuals who are immunosuppressed are also at risk of suffering severe infection and SSPE, likely owing to the high levels of virus reached during acute infection^{20,21}.

More generalized post-viral symptoms have also been reported, such as chronic fatigue syndrome/myalgic encephalomyelitis in a cohort who survived a pandemic H1N1/09 infection or persistent arthralgia in severe cases of CHIKV infection^{136,142}. Although the persistence of respiratory RNA viruses in the respiratory tract has long been recognized in individuals who are immunocompromised, numerous studies during the COVID-19 pandemic demonstrated that immunocompetent and otherwise healthy individuals can experience the

persistence of viral products^{100,101,143}. It is also established that severe infection with respiratory viruses like RSV or PIVs can predispose individuals to develop chronic respiratory conditions of the lung, including asthma and chronic obstructive pulmonary disease^{144,145}, and recent studies using the murine PIV SeV demonstrated a direct causal relationship between persistent PIV products and the development of asthma-like disease in mice^{2,144,145}.

Inflammation is a key component of the antiviral response, involving the activation of immune and non-immune cells and the production of proinflammatory cytokines to help control viral infections. However, when acute inflammatory responses fail to resolve the infection, sustained immune cell infiltration and cytokine production can promote tissue damage, remodeling and chronic symptoms. One major challenge in understanding the molecular mechanisms linking acute viral infection to chronic disease is the lack of animal models that accurately recapitulate human pathology. In addition, sensitive virus detection methods that could be applied to finding the rare virus-containing cells in human organs are lacking. The development of these tools or the use of representative models is therefore essential to advance our understanding of viral persistence and its long-term consequences. Using SeV as a model PIV, together with a battery of tools that eliminate concerns of false positives, including the use of humanized antibodies and transgenic reporter mice, our group demonstrated that the persistence of SeV RNA and protein is a major driver of chronic type 2 inflammation in the lung. This inflammation is like asthma in humans, showing a direct link between respiratory viral persistence and chronic inflammatory responses². Similarly, in vivo lineage tracing of CHIKV-infected cells identified myofibers, dermal and muscle fibroblasts and macrophages as harboring persistent CHIKV RNA at chronic time points, likely contributing to musculoskeletal inflammation associated with the chronic stage of CHIKV disease^{9,25,146}. Unresolved viral persistence may contribute to the onset of post-viral syndromes by directly or

indirectly disrupting tissue-specific immune responses. Developing more sensitive detection methods and better animal models that help elucidate the cellular and molecular basis of these syndromes is critical for the identification of therapeutic or diagnostic targets to mitigate the long-term burden of viral infections.

Conclusions

The traditional view of RNA viruses as strictly acute agents is being reshaped by emerging evidence of their capacity to persist as intact or mutated genomes, nsVGs or proteins found in diverse cellular reservoirs. These findings reveal that beyond the acute phase, common RNA viruses leave molecular footprints that have functional consequences in contributing to chronic inflammation, immune dysregulation and long-term organ dysfunction. The causal link between acute RNA virus persistence and post-viral sequelae should raise awareness of the long-term risks of what are usually assumed to be short-course viral infections. If these acute infections are severe, there is a significant chance that a chronic condition will manifest years later. Preventing severe acute infections in susceptible individuals through vaccination or isolation may diminish this risk^{147,148}.

Several questions remain unanswered. Can all acute RNA viruses persist? What are the mechanisms of viral RNA retention in the diverse cell reservoirs? What is the fate of cells that survive infection, and how are these surviving cells different from noninfected or bystander cells? What are the host cell molecular determinants that characterize a productive RNA virus infection versus a persistent infection? What is the exact role of the immune system in the development or resolution of chronic disease? It is also essential to investigate whether different RNA viruses or their products induce common transcriptomic or epigenomic reprogramming in their host cells and whether such footprints can serve as biomarkers for chronic inflammatory syndromes.

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